

Group theory

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Chapter 1. Groups

This material follows the standard treatment in a [1].

Exercise 1.1. Consider an exercise.

See [Exercise 1.0](#).

Exercise 1.2. funn

Solve [Exercise 1.1](#).

§1.1 Fields

We begin by establishing the properties of fields mapping vector components.

Definition 1.3. A function $T : V \rightarrow W$ between two vector spaces over the same field F is called a **linear transformation** if it satisfies:

1. Additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in V$
2. Homogeneity: $T(cv) = cT(v)$ for all $c \in F$ and $v \in V$

[Definition 1.2.](#)

Definition 1.4. ...

§1.2 Degree of an extension

Theorem 1.5. Let V and W be vector spaces, where V is finite-dimensional. If $T : V \rightarrow W$ is a linear map, then:

$$\dim(\text{null } T) + \dim(\text{range } T) = \dim V$$

§1.3 Exercises

Exercise 1.6. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find a structured matrix representation M_T and verify the output dimension explicitly:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Exercise 1.7 (Identity Dimension Mapping).

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + y, z)$. Find the dimension of the null space.

Solution:

By inspecting the matrix row parameters:

$$M_T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The rank is clearly 2 because the two rows are linearly independent. By Rank-Nullity, $\dim(\text{null } T) = 3 - 2 = 1$.

Note 1.8. Remember that every basis is linearly independent.

Warning 1.9. Do **not** confuse image and codomain.

Example 1.10. Let

$$T(x, y) = (x + y, y).$$

Proof. We proceed by induction...

□

Remark 1.11. The converse is false.

History 1.12. The Rank-Nullity theorem appeared in the nineteenth century.

Lemma 1.13. lemma statement

Proposition 1.14. proposition statement

Claim 1.15. claim statement

Corollary 1.16. corollary statement

This material follows the standard treatment in a [\[1\]](#).

Bibliography

[1] D. S. Dummit and R. M. Foote, *Abstract Algebra*, 3rd ed. Wiley, 2004.

Chapter 2. Symmetric groups

§2.1 eqn test

$$a = b. \tag{1}$$

Study the equation in [Equation 1](#).

$$c = d. \tag{2}$$

Study the equation in [Equation 2](#).

§2.2 Core Definitions

In this session, we analyze how linear transformations scale specific vector subspaces.

§2.2.1 some defn

Definition 2.1. Let $T : V \rightarrow V$ be a linear operator on a vector space V over a field F . A non-zero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar $\lambda \in F$ such that:

$$T(v) = \lambda v$$

The scalar λ is called the **eigenvalue** corresponding to the eigenvector v .

See [Definition 2.0](#) for the formal definition of eigenvalues and eigenvectors.

§2.3 Fundamental Theorems

Theorem 2.2. Let $T : V \rightarrow V$ be a linear operator. If $v_1, v_2, \dots, v_m$ are eigenvectors corresponding to **distinct** eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$, then the set $\{v_1, v_2, \dots, v_m\}$ is linearly independent.

By [Theorem 2.1](#), we conclude that eigenvectors corresponding to distinct eigenvalues are linearly independent.

§2.4 Homework Practice

Exercise 2.3. Consider the 2×2 matrix A representing a linear transformation on $\mathbb{R}^2$:

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

1. Compute the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.
2. Find the distinct eigenvalues λ_1 and λ_2 by solving:

$$\lambda^2 - 7\lambda + 10 = 0$$

By [Exercise 2.2](#).

Lemma 2.4. lemma

by [Lemma 2.3](#).

Chapter 3. Sylow's theorems

Chapter 4. Core Definitions

Theorem 3.1. There are infinitely many prime numbers.

Lemma 3.2. lemma nice

Note that [Lemma 3.1](#)

also Note that [Lemma 3.1](#)

Corollary 3.3. Let K/F be a field extension.

Note that [Lemma 3.1](#) [Corollary 3.2](#)

Theorem 3.4. nice thm

observe that [Lemma 3.1](#) [Corollary 3.2](#)